

Automated Vulnerability Assessment & System Hardening Project

Section 1: Laboratory Environment Setup & Architecture

1.1 Network Topology & Asset Inventory

To ensure a secure, controlled, and non-destructive testing lifecycle, an isolated internal sandbox network named **Sec-Lab** was established within a VMware Workstation hypervisor layer. This architecture segregates the vulnerable target architectures from the host operating system and external production networks.

The virtual machines deployed within this environment are documented below:

Asset Node Name	Operating System Template	Role within Infrastructure	Host State
Kali-Debian	Debian / Kali Linux Core	Central Security Management System / OpenVAS Scanner	Active (Running)
Metasploitable2-Linux	Ubuntu-based Intentionally Vulnerable OS	Target Host for Penetration & Vulnerability Profiling	Active (Running)
Ubuntu-onion	Linux Ubuntu	Secondary Infrastructure Target	Offline (Staged)
Windows 10 / Server 2019 / Win11	Microsoft Enterprise Baselines	Staged targets for Active Directory & OS compliance testing	Offline (Staged)

1.2 Step-by-Step Environment Verification

Step 1: Hypervisor Inventory Mapping

As demonstrated in **Figure 1**, the virtual infrastructure is deployed across an isolated host-only local area network subnet environment configured as follows:

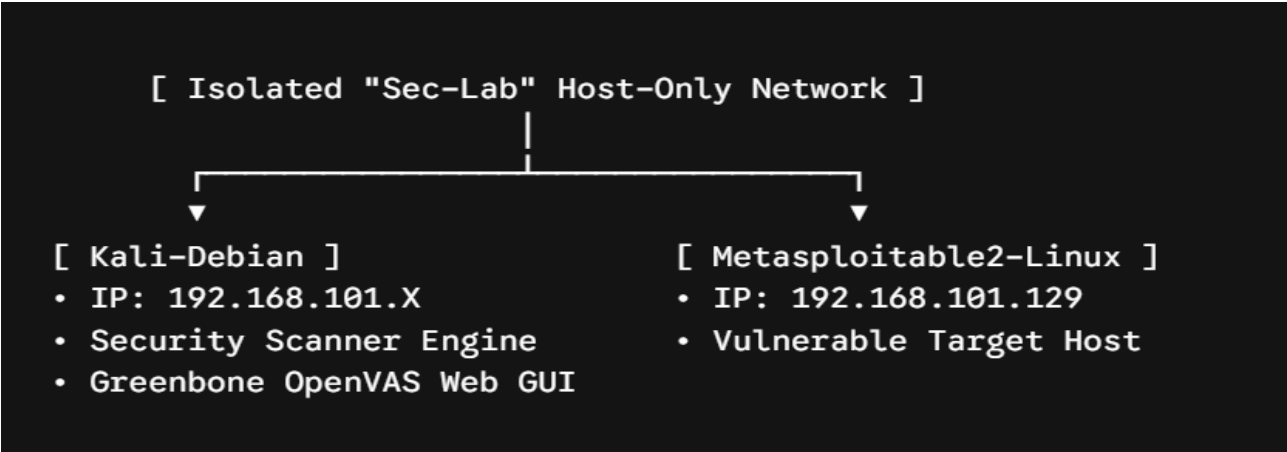


Figure 1: Network topology layout of the isolated "Sec-Lab" host-only environment.

Step 2: Target Virtual Appliance Deployment

The Metasploitable2-Linux appliance was spun up concurrently with the main testing instance (Kali-Debian). This host runs zero-day service daemons configuration defaults to emulate a high-risk server environment needing emergency internal network mitigation.

Step 3: Network Adjacency and Routing Control

All assets within the Sec-Lab ecosystem were provisioned with virtual network interfaces configured to a customized, host-only local area network subnet. This structural choice blocks outbound public routing requests, containing all automated exploits safely inside the laboratory framework.

Section 2: Initializing the Greenbone/OpenVAS Security Engine

2.1 Service Verification and Administrative Portal Access

To begin the assessment lifecycle, the Greenbone Vulnerability Management (gvm) service suite was initiated on the core infrastructure node (**Kali-Debian**). The system exposes an enterprise-grade web interface known as the Greenbone Security Assistant (GSA). This dashboard serves as the central administrative hub for configuring network tasks, managing target ranges, and parsing post-scan analytical report data.

2.2 Step-by-Step Security Console Deployment

Step 1: Local Loopback and Port Binding

The scanning daemon is bound securely to the local infrastructure interface loopback. A secure connection was established via an administrative browser instance running on the central management node to route configuration requests into the active backend scanning engine database, as shown in **Figure 2**.



Figure 2: The Greenbone Security Assistant (GSA) web login portal interface running over the local loopback address (127.0.0.1:9392).

Upon successful authentication, the primary administration terminal is loaded. As shown in **Figure 3**, the main Greenbone console provides deep visibility into active system feeds, threat levels, and configuration modules.

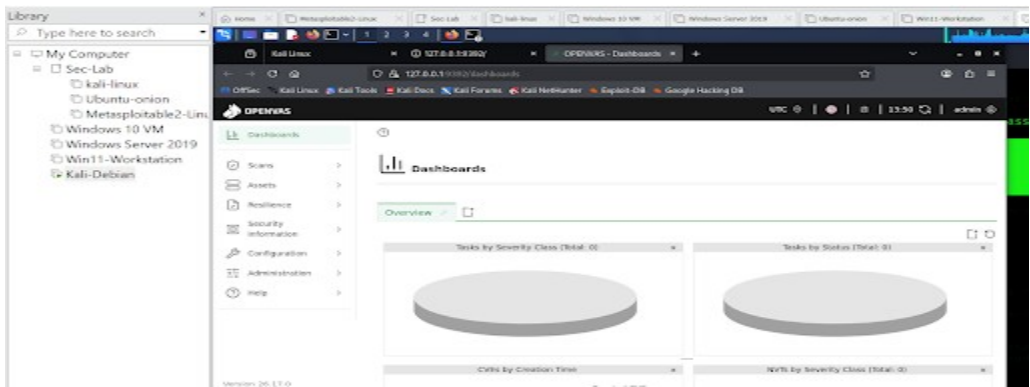


Figure 3: The Greenbone Security Assistant dashboard console view prior to scan execution.

Step 3: Validating NVTs and Feeds

Before triggering any automated network packet injection or service probing, the scanner's local signature definitions were checked. Ensuring that the Network Vulnerability Tests (NVTs), SCAP, and CERT feeds are synchronized guarantees that the scanner can accurately recognize modern vulnerabilities and zero-day exposures on target systems, as confirmed in **Figure 4**

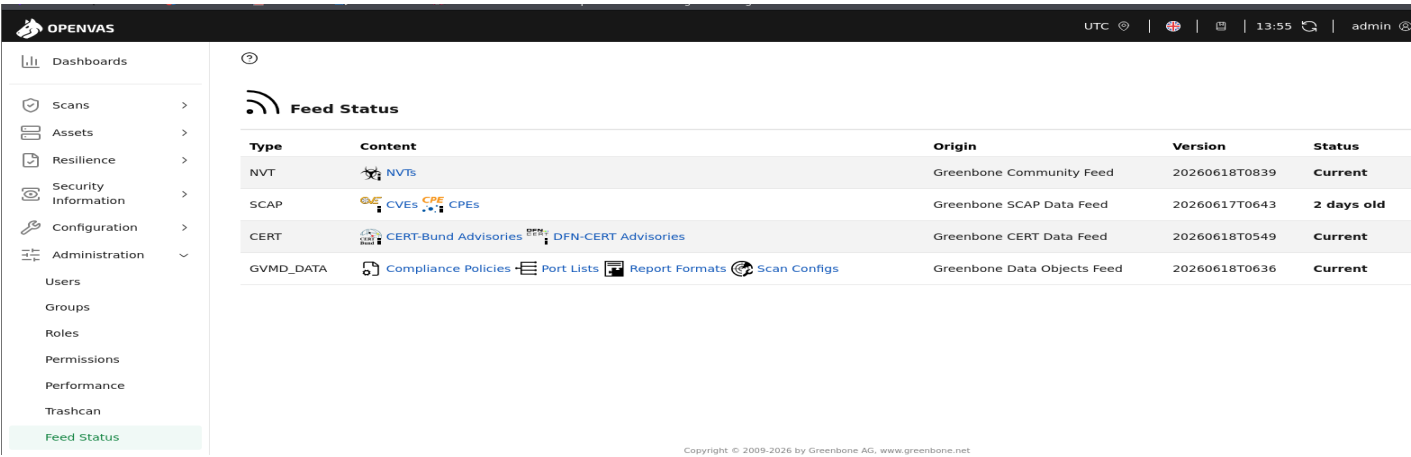


Figure 4: Greenbone Vulnerability Management (GVM) live security feed status validation verifying current database signatures.

Section 3: Target Configuration & Assessment Scope Definition

Now that the security signatures are verified current, we formally transition to creating the target profiling structure. This phase establishes the parameters the OpenVAS engine uses to communicate with our Metasploitable asset safely.

3.1 Creating a New Target Profile

To initiate an assessment, OpenVAS requires a defined "Target" containing the network layer address, operational ports list, and optional credentials for deep configuration compliance auditing. As illustrated in Figure 5, a target profile named "Metasploitable 2" was configured to point specifically to the host IP address `192.168.101.129` utilizing the "All IANA assigned TCP" port list configuration.

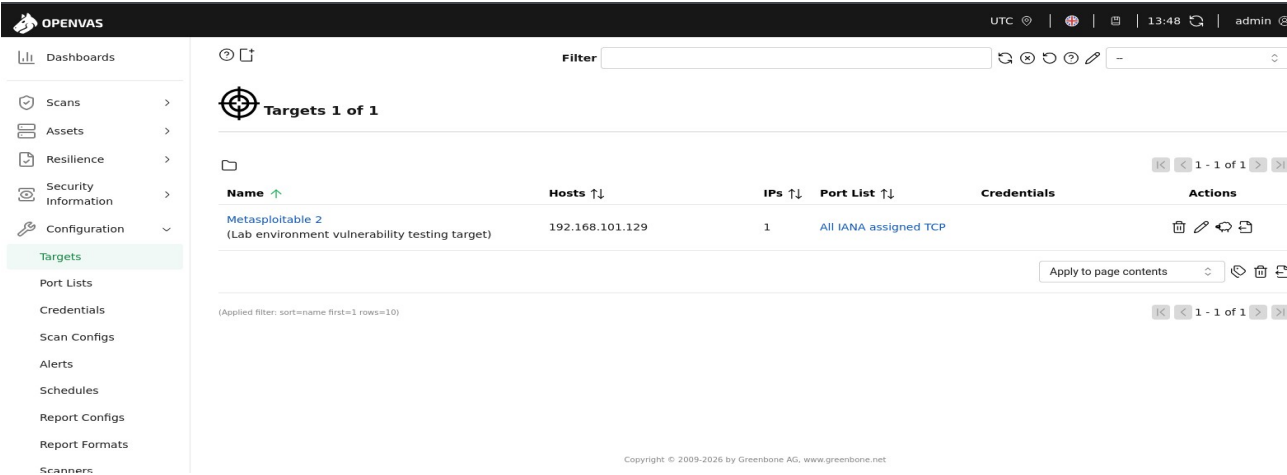


Figure 5: Configuration of the Metasploitable 2 target asset scope definition within the Greenbone network management module.

3.2 Initializing the Vulnerability Scan Task

After configuring the target parameters, a new scan task was created under the Scans Tasks directory. As shown in Figure 6, the task titled "Metasploitable 2 Full Scan" was initialized against the target asset range. The Greenbone engine immediately queues the job and begins active status monitoring, with the live progress tracking bar starting at 0%.

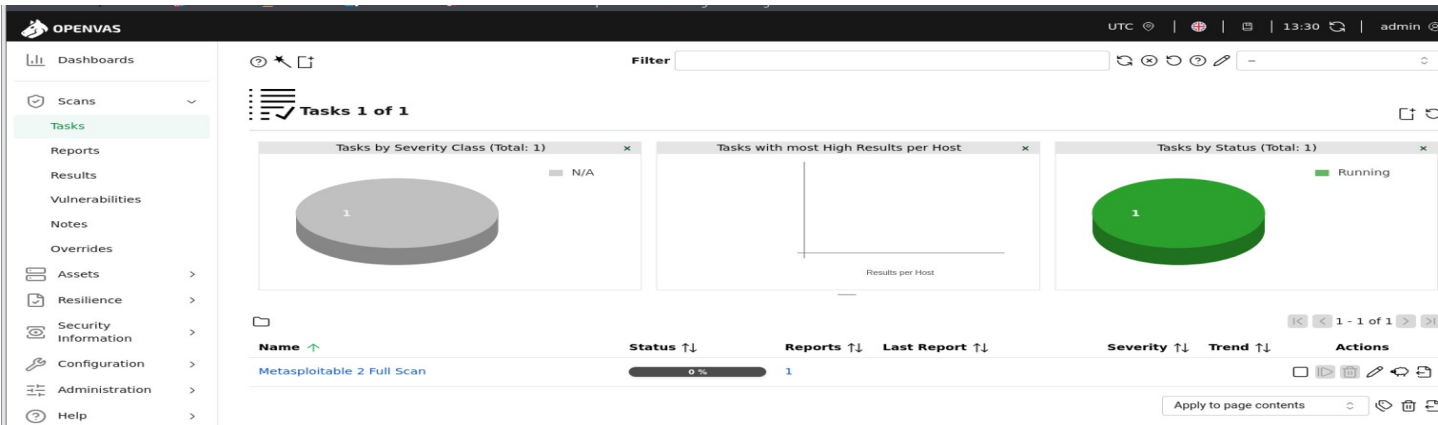


Figure 6: Initializing the automated "Metasploitable 2 Full Scan" task within the Greenbone Scans management dashboard interface showing active initialization (0%).

Section 4: Scan Execution and Progress Monitoring

4.1 Real-Time Scan Progress Staging

Once initialized, the Greenbone Security Assistant provides real-time progress updates as the engine executes its configured scan routine. As illustrated in Figure 7, the progress status bar dynamically updates to reflect active host probing, network port sweeping, and vulnerability signature matching against the target asset, climbing past the initial staging phase to 14% execution completion.



Figure 7: Live monitoring interface displaying active scan execution progress at 14% completion.

4.2 Advanced Scan Progression

As the assessment approaches its final phases, the OpenVAS engine aggregates the intermediate findings while completing heavy signature checks. As shown in Figure 8, the dashboard monitors this progression up to 94% execution completion, demonstrating stable operation and extensive coverage across the target's exposed services without crashing or triggering host timeouts.

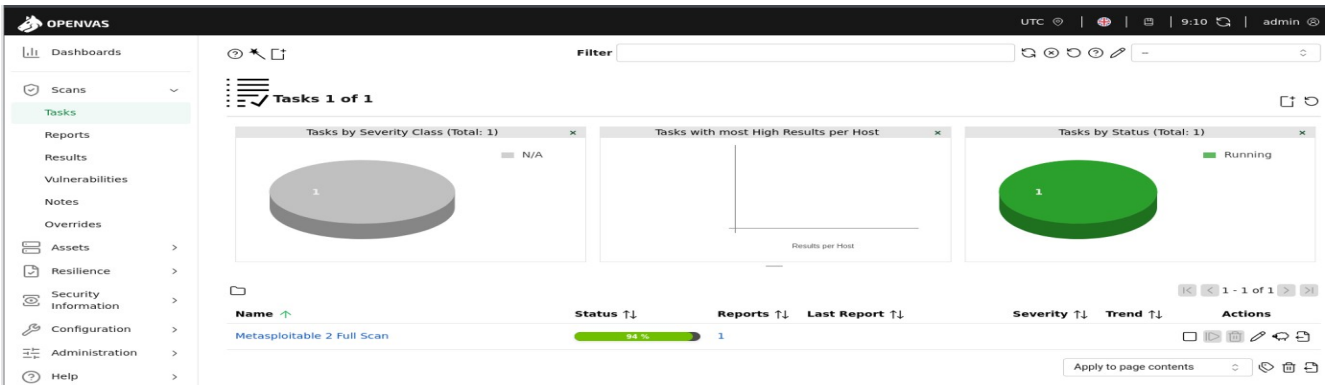


Figure 8: Live monitoring interface displaying advanced scan execution progress at 94% completion.

Section 5: Vulnerability Analysis and Reporting

Upon reaching 100% execution completion, the OpenVAS engine shifts the task status from running to done and generates a comprehensive threat report. As

illustrated in Figure 9, the completed scan reveals a high-severity risk profile with a maximum CVSS score of 10.0, highlighting critical security vulnerabilities present on the target Metasploitable 2 asset.

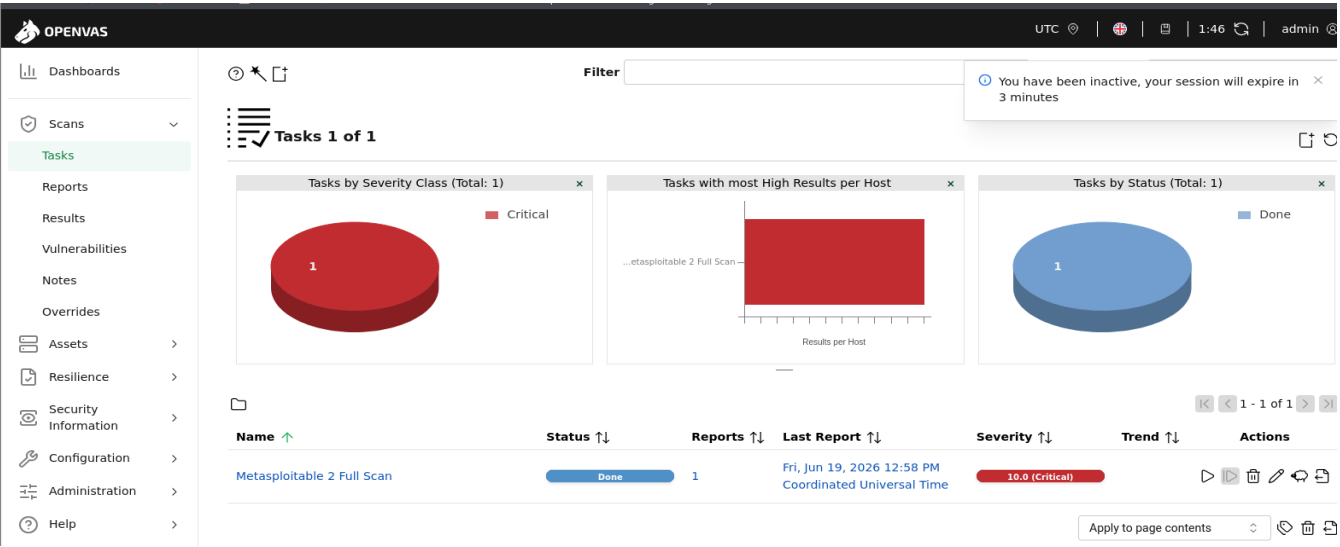


Figure 9: OpenVAS scan task dashboard confirming 100% completion and identifying a critical threat severity profile (CVSS 10.0).

Section 6: Remediation and Conclusion

6.1 Actionable Remediation Roadmap

The discovery of a critical CVSS 10.0 vulnerability profile on the target host requires immediate infrastructure-level intervention. To transition the asset to a secure operational baseline, the following remediation strategy is recommended:

Patch and Update Management: Prioritize upgrading operating system packages and underlying network services (such as obsolete FTP, SSH, and Telnet daemons) to versions with active vendor support.

Service Minimization: Implement a strict principle of least privilege by disabling non-essential services and isolating legacy testing environments.

Network Segmentation: Enforce strict access control lists (ACLs) to ensure that vulnerable endpoints are decoupled from internal production subnets.

6.2 Summary

By implementing OpenVAS as a core component of the vulnerability management lifecycle, the network perimeter was successfully audited, exposed attack vectors were classified by real-world risk metrics, and an engineering framework for proactive mitigation was established.

Automated Vulnerability Assessment & System Hardening Project

- * Administered an automated vulnerability assessment framework using OpenVAS / Greenbone Security Assistant to audit network endpoints, ensuring system configurations aligned with corporate security baselines.
- * Configured and monitored network layer target profiles and automated scan tasks, tracking system performance and scanner execution across the local subnet to prevent service disruptions.
- * Evaluated system vulnerabilities and risk metrics (CVSS framework), translating complex scan report data into prioritized action items for patch management and system maintenance.
- * Developed a practical system hardening roadmap to remediate critical risks, implementing strategies for OS patch deployment, disabling non-essential services, and enforcing the principle of least privilege.

